Unreasonable Confidence from Simulator Misspecification: Directional Robustness on the SPEED+ Sim-to-Real Benchmark

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Abstract

Vision-based navigation systems for orbit must be trained almost entirely on synthetic imagery, because real labelled images of a given spacecraft in flight rarely exist. Benchmarks like SPEED+ are built for exactly that situation: the test sets are real images of a satellite mockup under two lighting conditions, but the training set comes from a simulator. That simulator approximates how light reflects off mylar and aluminium, and like every scientific simulator it is misspecified. The error is radiometric: it does not reproduce the specular highlights, hard shadows, and sensor saturation that are ordinary in real images. The resulting synthetic-to-real gap is the central risk for such systems. We probe it with a deliberately simple monocular pose estimator—a 2×2 study of backbone adaptation and photometric augmentation, trained on synthetic images and tested on both real hardware-in-the-loop domains. We report three findings. (1) The model does not know it is failing. Regions conformally calibrated to 90% on synthetic data cover 9–15% of rotation errors on sunlamp and 21–38% on lightbox, so a wrong pose carries the confidence of a correct one. (2) Transfer is asymmetric. Scaling data and features improved synthetic rotation error from 1.67 to 1.05 rad but did not detectably move the harshly lit *sunlamp* domain (0.03 rad, within seed variation), whereas adapting the backbone's last stage carried 70–80% of its marginal synthetic gain across to both real domains. (3) Once the backbone adapts, augmentation only helps in the direction it models. Our photometric stack improved the diffusely lit *lightbox* domain from 1.17 to 0.99 rad on every seed, yet moved sunlamp no further than seed-to-seed variation; with the backbone frozen, it helped sunlamp more.

1 Introduction

Machine learning models for spacecraft navigation are trained almost entirely on the output of a simulator and deployed against a physical system that simulator only approximates. Real images of a target spacecraft in orbit, labelled with its pose, essentially do not exist, so estimators learn from rendered images and are then tested for the first time during the mission. The SPEED+ benchmark [10] was built to make that discrepancy measurable. It pairs $\sim$60,000 labelled synthetic training images of the Tango spacecraft with $\sim$9,500 real images of a physical mockup, lit two ways: *lightbox*, diffuse and Earth-albedo-like, and *sunlamp*, a collimated solar simulator giving harsh directional light. On its predecessor SPEED, whose real test images show the same mockup, the 2019 challenge's top ten entries degraded 2.7–40$\times$ in pose score from synthetic to real (winner 40$\times$, runner-up 5.3$\times$, median 8.5$\times$) [6].

The simulator is not a physics engine. SPEED+'s synthetic images come from an OpenGL rasteriser—the Optical Stimulator camera emulator—whose shading is an empirical model of how sunlit foil and metal look, chosen for speed over fidelity [10]. That shading model is a closure in the same

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sense as an empirical force field in molecular dynamics or a sub-grid parametrisation in a climate model: a cheap stand-in for physics the simulator never resolves, and the place its error lives. The error is radiometric, not geometric. Ranges and pose distributions match across domains; what differs is appearance. Mylar and machined metal reflect specularly, and a collimated source turns those reflections into hard shadow boundaries and saturated pixels. Saturation is not even a shading error—it happens in the detector, so the observation operator is misspecified as well as the scene. A photometric augmentation stack is therefore a *discrepancy model*: an unvalidated guess at the simulator-to-reality residual, applied not as a correction to the forward model but as a distribution the training data is smeared over. This paper is about what happens when that guess is wrong—not only a loss of accuracy, but the silent loss of calibration.

We use SPEED+ as a controlled testbed and ask *what actually transfers from simulation to reality*. The estimator is a direct-regression baseline: an ImageNet-pretrained ResNet-50 with a small head. We train it on the synthetic split alone, in a 2×2 design of {frozen backbone, last stage unfrozen} $\times$ {no augmentation, photometric augmentation}, and evaluate on the full *lightbox* and *sunlamp* test sets. Three of the four arms are run on three seeds, and checkpoints are selected on synthetic validation only. Our contributions:

- **The model does not know it is failing.** Split-conformal regions calibrated to 90% on held-out synthetic data keep that coverage in-domain, but cover only 9–15% of rotation errors on sunlamp, and test-time BatchNorm adaptation lifts that only to 11–20%. On the weakest arms the attitude cone spans roughly half of all orientations: still valid, no longer meaningful.
- **Transfer is asymmetric.** Scaling data and features under a frozen backbone bought large synthetic gains that barely crossed to harsh illumination: 0.61 rad of synthetic improvement left sunlamp within seed variation, whichever factor was scaled. Gains from adapting the representation crossed substantially.
- **Augmentation is directional.** Our photometric stack was built in good faith to anticipate harsh real illumination, but it models neither real domain well, and the *diffuse* lightbox markedly less badly: adapting layer 4, it improved lightbox by $\sim$0.18 rad on every seed, while sunlamp moved no further than seed-to-seed variation; with the backbone frozen, the order reverses. Ablating its components does not resolve sunlamp either (Fig. 1).

2 Setup

Data. From the SPEED+ synthetic training split we sample 28,000 training and 2,000 validation images, downscaled to 320×512 grayscale to fit the GPU available. Both real domains are used in full as test sets: lightbox (6,740 images) and sunlamp (2,791). No real image influences training, tuning, or model selection. Range distributions match across domains, so any extra difficulty on the real domains comes from appearance rather than from harder poses.

Model and loss. A ResNet-50 [4] pretrained on ImageNet [3] feeds global-average-pooled 2048-d features to a two-layer MLP head that regresses translation $\mathbf{t}$ (targets z-scored per component) and a unit quaternion $\mathbf{q}$—direct pose regression [5, 13, 12]. The loss splits the two: $\mathcal{L} = \mathrm{MSE}(\hat{\mathbf{t}}, \mathbf{t}) + \lambda\,(1 - \langle\hat{\mathbf{q}}, \mathbf{q}\rangle^2)$ with λ=1 and $\hat{\mathbf{q}}$ ℓ_2-normalised. The squared inner product is sign-invariant, and labels are canonicalised to one hemisphere. Together these remove the double-cover ambiguity—$\mathbf{q}$ and $-\mathbf{q}$ encode the same rotation— which otherwise gives naive quaternion MSE contradictory targets [17].

The 2×2 arms. In the *"frozen"* arms the backbone runs in inference mode and keeps its ImageNet BatchNorm statistics, so only the head is trained. In the *"unfrozen"* arms the final ResNet stage (layer 4) is trained as well, so its BatchNorm statistics track the (augmented) synthetic training distribution. All arms share one configuration (Adam, learning rate 10^{-3}, batch 16, 20 epochs, checkpoints on even epochs), and each arm keeps the checkpoint with the best *synthetic* validation rotation error. Three arms are repeated over three seeds; the frozen un-augmented arm is single-seed, with head-only retrains in App. D.

Augmentation. The photometric factor is a good-faith attempt to anticipate harsh real illumination: brightness (0.65–1.5) and contrast ($\pm$0.5) jitter, log-uniform gamma in $[0.5, 2]$, a soft clamped additive

Table 1: Full results: mean ± s.d. over seeds {42, 43, 44}; the frozen un-augmented arm is single-seed (†). E_T range-normalised, E_R in rad (chance level 2.207); checkpoints picked by best synthetic val. E_R.

		Synthetic (val)		Lightbox		Sunlamp	
Backbone	Aug.	E_T	E_R	E_T	E_R	E_T	E_R
frozen†	–	0.066	1.053	0.151	1.752	0.253	2.048
frozen‡	✓	0.073±0.002	1.185±0.019	0.121±0.005	1.708±0.031	0.304±0.024	1.876±0.010
layer 4	–	0.036±0.002	0.317±0.010	0.169±0.027	1.166±0.018	0.311±0.036	1.534±0.075
layer 4	✓	0.042±0.002	0.322±0.027	0.150±0.017	**0.987±0.010**	0.355±0.081	1.468±0.119

†n=1 (seed 42); 3-seed head-only retrains: E_R s.d. 0.017, 0.026, 0.031. ‡Seeds 43, 44 on an Apple M5. Per-seed rows: App. A.

Gaussian "sun glare" blob ($\sigma \in [10, 50]$ px), Gaussian blur, and additive Gaussian noise (stochastic components at p=0.5), plus random erasing. Section 3.2 examines what it actually prepares the model for.

Metrics. We track normalised translation error $E_T = \|\hat{\mathbf{t}} - \mathbf{t}\|/\|\mathbf{t}\|$ and geodesic rotation error $E_R = 2\arccos|\langle\hat{\mathbf{q}}, \mathbf{q}\rangle|$. Their sum is the official SPEED+ score $S = E_T + E_R$ [6]; we report the unthresholded sum, whereas SPEC2021 [11] additionally zeroes errors below a calibration threshold. Under uniformly random orientations the rotation angle has density $(1-\cos\theta)/\pi$, so random guessing achieves $\mathbb{E}[\theta] = \pi/2 + 2/\pi \approx 2.207$ rad; we call this the *chance level*.[1] We define *skill* as $2.207 - E_R$, which separates errors that look close. As raw errors, 1.67 and 2.07 rad are neighbours; as skill they are 0.54 and 0.13 rad—meaningfully better than chance, and barely better than chance.

Code and data. SPEED+ comes from its official hosts. The code, configs, per-seed results, and conformal analysis behind every number here are at `https://github.com/lokindoo/spacecraft-pose-estimation`.

3 Results

3.1 Synthetic gains transfer asymmetrically

Our initial frozen-backbone configuration (4,000 training images, the head reading the backbone's 1,000-d ImageNet logits rather than pooled features) reached $E_R = 1.666$ rad on the same 2,000-image synthetic validation split used throughout, 2.002 on lightbox, and 2.073 on sunlamp. Scaling to 28,000 images and 2048-d pooled features improved synthetic rotation by 0.613 rad (1.666 → 1.053), but only 0.025 rad on sunlamp (2.073 → 2.048), less than the frozen head's seed s.d. there (0.031 rad). Lightbox absorbed roughly 40% of the gain (2.002 → 1.752). Under harsh illumination the improvement all but vanished: the scaled baseline holds 1.15 rad of synthetic skill but keeps only 14% of it on sunlamp (39% on lightbox).

Pre-registered checks (App. D) support this reading. Over three seeds, the four {4k, 28k} × {logits, pooled} cells split the synthetic gain roughly evenly between data and features (0.33 and 0.29 rad), and neither moves sunlamp beyond seed variation. A tuned head gains another 0.10 rad on synthetic and nothing on the real domains. Re-estimating all BatchNorm statistics on synthetic images, weights fixed, makes the frozen head worse everywhere, so the layer-4 gain below is not a matter of normalisation statistics alone.

Unfreezing layer 4 behaves differently (Table 1). Synthetic E_R drops 1.05 → 0.32, and this time the real domains follow: sunlamp 2.05 → 1.53 and lightbox 1.75 → 1.17, so roughly 70–80% of the marginal synthetic gain carries across. What produced a gain therefore matters as much as its size: showing the frozen ImageNet backbone more data helped little, while adapting the representation helped a lot. Even so, the layer-4 no-augmentation arm keeps only 36% of its skill on sunlamp (55% on lightbox), and its SPEED+ score degrades 3.8–5.2× (S=0.353 synthetic, 1.335 lightbox, 1.845 sunlamp), at the low end of the 2019 challenge's range but not evidence of a smaller gap (Sec. 4).

[1] SPEED+ orientations are not uniform on $SO(3)$, so we also check the level empirically. A single constant pose fitted on the 28,000 training labels alone—never on a test domain—scores $E_R = 2.176$ on synthetic val, 2.203 on lightbox and 2.199 on sunlamp ($E_T = 0.415, 0.367, 0.372$), i.e. at most 0.031 rad from the uniform value, so we use 2.207 throughout. These are achievable scores of a fitted constant, not of a provably optimal one, and therefore upper-bound the best constant predictor. App. C gives the method.

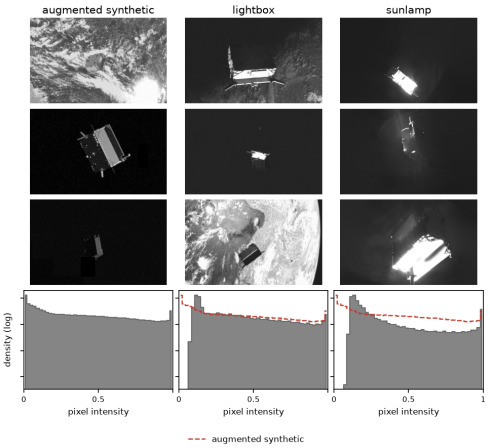


Figure 1: What the augmentation actually models. Columns: augmented synthetic training samples, and real test samples from each domain. Below each is the pixel-intensity density (log counts), pooled over every image in that split. The augmented density is overlaid on both real columns (dashed): far from both, and closer to lightbox—total variation 0.63 against 0.80. Gradient statistics give the same ordering (App. D).

3.2 Augmentation is directional

Seeds are paired across arms (42, 43, 44 in both), so we read the per-seed differences rather than comparing spreads. On lightbox, augmentation improves every seed: -0.183, -0.185, -0.170 rad, i.e. 3/3 at $\sim$15%, moving the mean from 1.166 ± 0.018 to 0.987 ± 0.010 rad. On sunlamp the paired differences change sign: $+0.116$, -0.265, -0.047 rad ($1.534 \pm 0.075 \rightarrow 1.468 \pm 0.119$), a spread wider than the effect it would have to establish. Translation is a weaker indicator, but it points the same way (Table 1): lightbox improves within noise, while on sunlamp two of the three augmented seeds (0.405, 0.399) fail to beat the fitted constant's 0.372. Augmentation pushes sunlamp translation to the no-skill line. On the unfrozen arms the in-domain synthetic error is essentially unchanged ($0.317 \rightarrow 0.322$ rad), so the lightbox gain does not come at the cost of the synthetic reference. We report absolute error, not the synthetic-to-real "gap", which can narrow merely because the reference degrades: with the backbone frozen, augmentation worsens synthetic error on every seed (by 0.134–0.169 rad, paired against head-only retrains; App. D). There the direction also reverses: sunlamp improves on every seed (-0.069 to -0.132 rad), and by more than lightbox (-0.012 to -0.117), so augmentation favours lightbox only once layer 4 adapts.

Ablation (pre-registered; App. D). Dropping blur and noise lowers sunlamp E_R on two of three seeds (-0.184, $+0.006$, -0.117 rad) and worsens lightbox on all three; also dropping the glare blob recovers lightbox but not sunlamp. Neither sunlamp effect is resolved, and a same-hardware control rerun shifts seed 42's sunlamp error by -0.142 rad, as much as the effects.

Mechanism. The stack models *diffuse* illumination variation: jitter and a soft clamped blob simulate gentle, spatially smooth lighting change, which is what lightbox is. Sunlamp is not: a collimated beam produces highlights off metallic and foil surfaces, cast shadows, and *clipped* regions. Photometric jitter only brightens smoothly: the augmented images end up brighter overall than either real domain, yet clip less often than sunlamp (2.0% of pixels against 2.8%). We expected blur, noise and the glare blob to work against sunlamp's sharp, bright structure; the ablation does not confirm this, and per-pixel gradients do not show sunlamp as sharper, since most of its frame is dark (edge density 2.0%, against 6.6% for lightbox and 10.7% for the augmented stack). Gradients do show direction: un-augmented synthetic images are closer to lightbox than to sunlamp (gradient total variation 0.07 against 0.26), and the stack, mostly through blur and noise, pushes them away from both (0.29 and 0.51). Finally, layer 4's BatchNorm statistics track the augmented synthetic distribution, which fits neither real domain well and lightbox less badly (Fig. 1). Augmentation pushed training toward the domain it was already least wrong about. Domain randomization [15] helps where it brackets the target domain; ours brackets neither, and misses sunlamp by more.

3.3 The model does not know it is failing

Does the estimator's uncertainty survive the shift? We calibrate split conformal regions [16, 7, 1] per arm on 1,000 fresh synthetic images, used neither for training nor for checkpoint selection. The score is Euclidean metres for translation (a keep-out sphere) and geodesic angle for rotation (a cone), with the finite-sample-corrected quantile. At nominal 90%, in-domain test coverage is close to nominal for every arm and metric (87.7–91.4%); its Clopper–Pearson [2] intervals include 90% in every

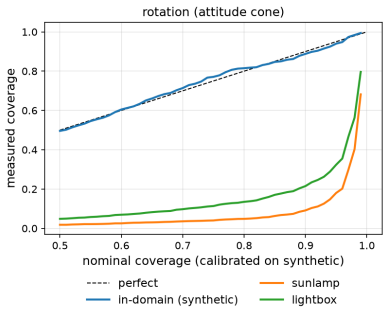


Figure 2: Conformal reliability of the attitude cone, layer 4 no-augmentation arm (seed 42). In-domain coverage tracks the diagonal; both real domains collapse far below it. The guarantee is intact—exchangeability is what broke.

case but one, the frozen no-augmentation arm's translation (87.7%, 85.5–89.7%), which is slightly anti-conservative. On the real domains it collapses. For the seeded no-augmentation arms, rotation coverage is 9–13% on sunlamp and 21–27% on lightbox (translation 14–18% / 33–35%); App. B gives every arm, domain and seed, with Clopper–Pearson intervals and inflation factors. Restoring 90% rotation coverage on sunlamp would mean inflating these arms' attitude cone $\sim$5–6$\times$, from $\sim$0.5 to $\sim$2.8 rad—a cone holding roughly 78% of all orientations.[2] Coverage mirrors the directional finding of Sec. 3.2, where augmentation increases lightbox rotation coverage to 33–38% but sunlamp only to 13–15%. The frozen arms fail the opposite way: their 90% attitude cone already spans 2.30–2.42 rad—49–56% of *all possible orientations*—valid in-domain, but operationally meaningless. A navigation filter fed these poses in deployment would receive wrong inputs that look exactly as confident as correct ones (Fig. 2). Test-time adaptation does not rescue this: re-estimating layer 4's BatchNorm statistics on unlabelled real images (AdaBN [8]; App. D) lowers rotation error on every layer-4 checkpoint (on average by 0.09 rad on lightbox, 0.16 on sunlamp), yet rotation coverage reaches only 24–40% and 11–20%.

4 Discussion and limitations

Ratios flatter weak models. A degradation ratio divides by in-domain error: on sunlamp even the skill-free constant pose (App. C) scores only 7.3$\times$ our synthetic S, so our modest ratio does not mean a small gap. Nor do the 2019 results separate method classes: PnP entries in the top ten degraded 4.8–40$\times$, the others 2.7–24$\times$ [6]. SPNv2 [9], the reference sim-to-real method on SPEED+, closes far more of the gap with multi-task representations and heavy style augmentation; we characterise how gains transfer under a simple estimator rather than compete with it.

Limitations. The frozen un-augmented arm was trained end to end on one seed, the ablation arms and two frozen augmented seeds on different hardware from the published ones, and sunlamp is the smaller test set (2,791 images): three seeds do not resolve sunlamp effects of 0.1 rad. We study one estimator on one benchmark, at 320$\times$512; with real-domain errors near 1 rad this is a diagnostic baseline, not a deployable navigator. Our directionality claim is about the match between an augmentation and the shift it faces, not about photometric augmentation in general, and our image statistics pool whole frames, which dark background dominates; object-masked statistics would test the mechanism more directly. Our conformal coverage is also marginal rather than conditional: fixed metre-radius spheres under-cover distant targets even in-domain, and range-normalising overcorrects.

Future work. The mechanism suggests its own tests: augmentation modelling sunlamp's actual structure (specular highlights, hard shadows, saturation clipping), full fine-tuning, and weighted conformal prediction [14], which needs density ratios between domains that barely overlap—exactly what sim-to-real lacks.

For fields that must rely on synthetic data, ask of a robustness intervention not "how much does it help?" but "which way does it push the training distribution?"; and validate uncertainty under the shift, or a wrong pose arrives wrapped in a confident interval.

[2]The fraction of $SO(3)$ lying within geodesic angle α of a fixed rotation is $F(\alpha) = (\alpha - \sin\alpha)/\pi$; so $F(2.79) \approx 0.78$, and $F(2.30)$ to $F(2.42)$ span 0.49 to 0.56.

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   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

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of translation targets, and the seeds $\{42, 43, 44\}$. Model selection is by best *synthetic* validation rotation error, applied uniformly across all arms. Hyperparameters were not tuned per arm: the configuration is held fixed across the whole 2×2 design, so the comparison is like-for-like. No real image influenced training, tuning, or model selection.

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   Justification: Three of the four arms are run on three seeds and reported as mean $\pm$ one standard deviation in Table 1, with all 30 per-seed values in App. A. Because seeds are paired across arms, Section 3.2 reads seed-paired differences rather than comparing spreads, and states the sign of every paired difference. Conformal coverage is reported with Clopper–Pearson binomial confidence intervals [2]. The frozen un-augmented arm was trained end to end on one seed, marked † in Table 1; three-seed head-only retrains give its spread (App. D), and the ablation arms are read as seed-paired differences against a same-hardware control. The variability captured is over the random seed (head initialisation, data ordering, and augmentation sampling).

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   Answer: [Yes]

   Justification: All training was done on a single cloud-hosted consumer-class GPU (Google Colab), which is why images are downscaled to 320×512 grayscale (Section 2). A 20-epoch run takes 320–355 s per epoch without augmentation and 780–795 s per epoch with it, i.e. about 1.8–2.0 h and 4.3–4.4 h per run: the augmentation pipeline dominates, and frozen and

unfrozen arms differ by under 10%. The eight Colab training runs behind Table 1 (four augmented, four not) total roughly 25 GPU-hours. Evaluation and conformal calibration are inference-only and take a few minutes per checkpoint. The revision experiments (App. D) ran on an Apple M5 laptop GPU: each of the seven layer-4 runs took about 3.7 h (about 26 h in total), the two frozen augmented runs in Table 1 about 3.2 h each, the cached-feature heads a few minutes each on CPU, and AdaBN and the image statistics under an hour together.

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A Per-seed results

Table 2 lists every run behind Table 1: all 30 arm × seed × domain evaluations. The frozen unaugmented arm was run on seed 42 only, the other arms on seeds 42, 43 and 44; frozen augmented seeds 43 and 44 were trained on an Apple M5 (App. D). Each run keeps the checkpoint with the best *synthetic* validation rotation error, chosen without reference to either real domain.

Table 2: Per-seed results. E_T is range-normalised; E_R is geodesic rotation error in radians (chance level 2.207).

Backbone	Aug.	Seed	Domain	E_T	E_R
frozen	–	42	synthetic (val)	0.066	1.053
frozen	–	42	sunlamp	0.253	2.048
frozen	–	42	lightbox	0.151	1.752
frozen	✓	42	synthetic (val)	0.071	1.180
frozen	✓	42	sunlamp	0.306	1.864
frozen	✓	42	lightbox	0.118	1.718
frozen	✓	43	synthetic (val)	0.073	1.205
frozen	✓	43	sunlamp	0.328	1.883
frozen	✓	43	lightbox	0.126	1.734
frozen	✓	44	synthetic (val)	0.074	1.169
frozen	✓	44	sunlamp	0.279	1.881
frozen	✓	44	lightbox	0.118	1.673
layer 4	–	42	synthetic (val)	0.039	0.306
layer 4	–	42	sunlamp	0.299	1.476
layer 4	–	42	lightbox	0.200	1.169
layer 4	–	43	synthetic (val)	0.035	0.326
layer 4	–	43	sunlamp	0.352	1.619
layer 4	–	43	lightbox	0.152	1.182
layer 4	–	44	synthetic (val)	0.035	0.318
layer 4	–	44	sunlamp	0.283	1.506
layer 4	–	44	lightbox	0.154	1.147
layer 4	✓	42	synthetic (val)	0.041	0.352
layer 4	✓	42	sunlamp	0.405	1.592
layer 4	✓	42	lightbox	0.152	0.986
layer 4	✓	43	synthetic (val)	0.044	0.302
layer 4	✓	43	sunlamp	0.399	1.354
layer 4	✓	43	lightbox	0.165	0.997
layer 4	✓	44	synthetic (val)	0.042	0.311
layer 4	✓	44	sunlamp	0.261	1.459
layer 4	✓	44	lightbox	0.132	0.977

B Conformal coverage, all arms and domains

Split-conformal regions are calibrated per arm on 1,000 held-out synthetic images used neither for training nor for checkpoint selection, at nominal 90% coverage with the finite-sample-corrected quantile. Coverage carries Clopper–Pearson 95% intervals [2]. $\hat{q}$ is the calibrated region size: a keep-out sphere radius for translation, a cone half-angle for rotation. *Inflation* is the factor by which $\hat{q}$ would have to grow to restore 90% coverage on that domain, so it measures directly how far the calibration is wrong. It sits near 1 in-domain and reaches 1.2–13× on the real domains, worst on the sharper unfrozen arms. A low factor on the frozen arms is not good news: their cones are already wide enough in-domain to be uninformative. The "synthetic (selection split)" rows are the 2,000-image validation split used for checkpoint selection, kept as a second, not fully independent, in-domain control.

Table 3 uses absolute scores: Euclidean metres for translation, geodesic radians for rotation. Table 4 repeats the translation analysis under the range-normalised score reported in the body. The two disagree on in-domain conditional behaviour—fixed metre spheres under-cover distant targets, range-

normalising overcorrects at short range—but the collapse on the real domains is the same either way.

Table 3: Conformal coverage at nominal 90%, absolute scores. Coverage in % with Clopper–Pearson 95% intervals; $\hat{q}_T$ in metres, $\hat{q}_R$ in radians.

				Translation (sphere)			Rotation (cone)		
Backbone	Aug.	Seed	Domain	Cov.	$\hat{q}_T$	Infl.	Cov.	$\hat{q}_R$	Infl.
frozen	–	42	synthetic (val)	87.7 (85.5–89.7)	0.712	1.11	90.0 (88.0–91.8)	2.299	1.01
frozen	–	42	sunlamp	29.2 (27.5–30.9)	0.712	4.76	58.2 (56.3–60.0)	2.299	1.28
frozen	–	42	lightbox	52.1 (50.9–53.3)	0.712	2.48	71.0 (69.9–72.1)	2.299	1.24
frozen	–	42	synthetic (selection split)	88.0 (86.5–89.4)	0.712	1.08	90.5 (89.1–91.7)	2.299	0.99
frozen	✓	42	synthetic (val)	89.3 (87.2–91.1)	0.793	1.01	88.7 (86.6–90.6)	2.403	1.02
frozen	✓	42	sunlamp	26.3 (24.6–27.9)	0.793	5.36	71.0 (69.3–72.7)	2.403	1.20
frozen	✓	42	lightbox	68.3 (67.2–69.4)	0.793	1.84	75.1 (74.1–76.2)	2.403	1.19
frozen	✓	42	synthetic (selection split)	89.9 (88.5–91.2)	0.793	1.01	89.5 (88.0–90.8)	2.403	1.02
frozen	✓	43	synthetic (val)	89.7 (87.6–91.5)	0.828	1.01	90.9 (88.9–92.6)	2.415	0.98
frozen	✓	43	sunlamp	23.4 (21.8–25.0)	0.828	5.16	70.0 (68.3–71.7)	2.415	1.20
frozen	✓	43	lightbox	66.9 (65.7–68.0)	0.828	1.91	75.5 (74.4–76.5)	2.415	1.18
frozen	✓	43	synthetic (selection split)	89.6 (88.2–91.0)	0.828	1.01	88.9 (87.4–90.2)	2.415	1.03
frozen	✓	44	synthetic (val)	89.8 (87.8–91.6)	0.839	1.00	91.1 (89.2–92.8)	2.372	0.97
frozen	✓	44	sunlamp	30.1 (28.4–31.9)	0.839	4.69	69.8 (68.1–71.5)	2.372	1.21
frozen	✓	44	lightbox	71.1 (70.0–72.2)	0.839	1.72	76.3 (75.2–77.3)	2.372	1.18
frozen	✓	44	synthetic (selection split)	89.8 (88.4–91.1)	0.839	1.01	89.5 (88.1–90.8)	2.372	1.01
layer 4	–	42	synthetic (val)	89.3 (87.2–91.1)	0.410	1.04	88.8 (86.7–90.7)	0.460	1.06
layer 4	–	42	sunlamp	18.2 (16.8–19.6)	0.410	10.92	9.1 (8.1–10.3)	0.460	6.07
layer 4	–	42	lightbox	34.1 (32.9–35.2)	0.410	7.92	21.5 (20.5–22.5)	0.460	5.52
layer 4	–	42	synthetic (selection split)	89.6 (88.2–90.9)	0.410	1.01	87.2 (85.7–88.7)	0.460	1.10
layer 4	–	43	synthetic (val)	90.2 (88.2–92.0)	0.378	0.99	89.1 (87.0–91.0)	0.540	1.04
layer 4	–	43	sunlamp	13.8 (12.6–15.2)	0.378	13.42	9.0 (8.0–10.1)	0.540	5.23
layer 4	–	43	lightbox	35.4 (34.3–36.6)	0.378	6.41	26.7 (25.6–27.7)	0.540	4.76
layer 4	–	43	synthetic (selection split)	89.2 (87.8–90.5)	0.378	1.02	89.2 (87.8–90.5)	0.540	1.05
layer 4	–	44	synthetic (val)	89.9 (87.9–91.7)	0.368	1.01	89.7 (87.6–91.5)	0.526	1.02
layer 4	–	44	sunlamp	18.0 (16.6–19.5)	0.368	11.73	12.9 (11.7–14.2)	0.526	5.30
layer 4	–	44	lightbox	32.9 (31.8–34.1)	0.368	6.52	26.5 (25.4–27.5)	0.526	4.71
layer 4	–	44	synthetic (selection split)	88.4 (87.0–89.8)	0.368	1.05	90.5 (89.1–91.7)	0.526	0.99
layer 4	✓	42	synthetic (val)	89.8 (87.8–91.6)	0.425	1.01	91.2 (89.3–92.9)	0.576	0.97
layer 4	✓	42	sunlamp	10.0 (8.9–11.2)	0.425	12.79	12.6 (11.4–13.9)	0.576	4.88
layer 4	✓	42	lightbox	44.3 (43.1–45.5)	0.425	6.45	37.2 (36.1–38.4)	0.576	3.82
layer 4	✓	42	synthetic (selection split)	89.8 (88.3–91.0)	0.425	1.01	90.0 (88.7–91.3)	0.576	1.00
layer 4	✓	43	synthetic (val)	90.4 (88.4–92.2)	0.494	0.98	91.1 (89.2–92.8)	0.483	0.96
layer 4	✓	43	sunlamp	10.7 (9.6–11.9)	0.494	10.80	15.3 (14.0–16.7)	0.483	5.59
layer 4	✓	43	lightbox	47.8 (46.6–49.0)	0.494	6.15	32.6 (31.5–33.7)	0.483	4.85
layer 4	✓	43	synthetic (selection split)	91.3 (90.0–92.5)	0.494	0.95	89.8 (88.3–91.0)	0.483	1.01
layer 4	✓	44	synthetic (val)	91.0 (89.1–92.7)	0.460	0.98	91.4 (89.5–93.1)	0.544	0.92
layer 4	✓	44	sunlamp	24.4 (22.9–26.1)	0.460	9.21	15.4 (14.1–16.8)	0.544	5.03
layer 4	✓	44	lightbox	51.0 (49.8–52.2)	0.460	5.10	38.0 (36.9–39.2)	0.544	4.19
layer 4	✓	44	synthetic (selection split)	91.0 (89.6–92.2)	0.460	0.97	90.8 (89.4–92.0)	0.544	0.96

C Constant-predictor baseline

The chance level of $\pi/2 + 2/\pi \approx 2.207$ rad assumes orientations drawn uniformly from $SO(3)$, which SPEED+ poses are not. This appendix gives the empirical counterpart quoted in Section 2: the score of a single fixed pose. It anchors "near chance" to a predictor one could actually build, rather than to a distributional assumption.

The constant is fitted on the 28,000 training labels only. Fitting it on a test domain would make it an oracle rather than a baseline, so neither real domain enters the estimate. For rotation we take the principal eigenvector of $M = \sum_i \mathbf{q}_i \mathbf{q}_i^\top$, which minimises chordal distance and is invariant to the quaternion double cover. We then refine it with five Karcher iterations on the sphere: sign-align each $\mathbf{q}_i$ to the current mean, log-map into the tangent space, average, and exp-map back. The $SO(3)$ geodesic distance is twice the S^3 one, so iterating on the sphere targets the same minimiser as our reported metric. For translation we take the arithmetic mean of the training translations. This yields $\mathbf{q} = (-0.198, 0.034, -0.019, -0.979)$ up to sign and $\mathbf{t} = (-0.0001, 0.0026, 6.120)$ m.

Table 4: Conformal coverage at nominal 90%, translation scored with the range-normalised error E_T used in the body.

Backbone	Aug.	Seed	Domain	Coverage	$\hat{q}$ (relative)	Infl.
frozen	–	42	synthetic (val)	89.4 (87.3–91.2)	0.1189	1.01
frozen	–	42	sunlamp	21.7 (20.2–23.3)	0.1189	3.85
frozen	–	42	lightbox	49.6 (48.4–50.8)	0.1189	2.42
frozen	–	42	synthetic (selection split)	89.3 (87.9–90.7)	0.1189	1.03
frozen	✓	42	synthetic (val)	88.9 (86.8–90.8)	0.1253	1.05
frozen	✓	42	sunlamp	15.7 (14.3–17.1)	0.1253	4.20
frozen	✓	42	lightbox	66.2 (65.0–67.3)	0.1253	1.78
frozen	✓	42	synthetic (selection split)	89.6 (88.2–90.9)	0.1253	1.01
frozen	✓	43	synthetic (val)	86.2 (83.9–88.3)	0.1259	1.10
frozen	✓	43	sunlamp	13.4 (12.1–14.7)	0.1259	4.46
frozen	✓	43	lightbox	61.7 (60.5–62.9)	0.1259	1.94
frozen	✓	43	synthetic (selection split)	88.5 (87.0–89.9)	0.1259	1.04
frozen	✓	44	synthetic (val)	88.6 (86.5–90.5)	0.1306	1.04
frozen	✓	44	sunlamp	19.1 (17.7–20.6)	0.1306	3.82
frozen	✓	44	lightbox	67.8 (66.7–68.9)	0.1306	1.69
frozen	✓	44	synthetic (selection split)	89.5 (88.1–90.8)	0.1306	1.02
layer 4	–	42	synthetic (val)	91.0 (89.1–92.7)	0.0787	0.96
layer 4	–	42	sunlamp	13.5 (12.2–14.8)	0.0787	6.93
layer 4	–	42	lightbox	34.1 (32.9–35.2)	0.0787	6.07
layer 4	–	42	synthetic (selection split)	90.2 (88.8–91.5)	0.0787	0.99
layer 4	–	43	synthetic (val)	92.0 (90.1–93.6)	0.0739	0.93
layer 4	–	43	sunlamp	9.0 (8.0–10.2)	0.0739	8.07
layer 4	–	43	lightbox	35.5 (34.4–36.7)	0.0739	4.70
layer 4	–	43	synthetic (selection split)	91.3 (90.0–92.5)	0.0739	0.95
layer 4	–	44	synthetic (val)	89.7 (87.6–91.5)	0.0678	1.01
layer 4	–	44	sunlamp	12.0 (10.9–13.3)	0.0678	7.68
layer 4	–	44	lightbox	32.2 (31.1–33.3)	0.0678	5.11
layer 4	–	44	synthetic (selection split)	89.2 (87.8–90.6)	0.0678	1.03
layer 4	✓	42	synthetic (val)	88.8 (86.7–90.7)	0.0814	1.07
layer 4	✓	42	sunlamp	5.2 (4.4–6.1)	0.0814	7.99
layer 4	✓	42	lightbox	43.7 (42.5–44.9)	0.0814	4.58
layer 4	✓	42	synthetic (selection split)	90.5 (89.1–91.7)	0.0814	0.99
layer 4	✓	43	synthetic (val)	90.3 (88.3–92.1)	0.0911	1.00
layer 4	✓	43	sunlamp	5.7 (4.9–6.6)	0.0911	6.96
layer 4	✓	43	lightbox	46.3 (45.1–47.5)	0.0911	4.48
layer 4	✓	43	synthetic (selection split)	90.8 (89.5–92.1)	0.0911	0.98
layer 4	✓	44	synthetic (val)	90.5 (88.5–92.2)	0.0894	0.99
layer 4	✓	44	sunlamp	18.7 (17.2–20.2)	0.0894	5.74
layer 4	✓	44	lightbox	51.4 (50.2–52.6)	0.0894	3.60
layer 4	✓	44	synthetic (selection split)	91.0 (89.7–92.2)	0.0894	0.97

Two caveats belong with the numbers. First, the Karcher steps do not contract sharply here: the training orientations cover most of the sphere, so the Fréchet functional is close to flat and the mean is weakly identified. This does not affect the result— the chordal and Karcher estimates lie 0.022 rad apart and agree to 0.001 on every score in Table 5. Second, both estimators minimise *squared* error, whereas we report mean unsquared error, whose minimisers are the corresponding geometric medians. The scores below are therefore achievable rather than optimal, and so upper-bound the best constant predictor. We do not claim optimality, and no conclusion in the paper needs it.

Table 5: A constant pose fitted on the training split, evaluated on each test domain. E_T range-normalised, E_R in radians. Skill is $2.207 - E_R$. Reported for the Karcher-refined estimate.

Domain	n	E_T	E_R	Skill
synthetic (val)	2,000	0.415	2.176	0.031
lightbox	6,740	0.367	2.203	0.005
sunlamp	2,791	0.372	2.199	0.009

The empirical no-skill rotation level therefore sits at most 0.031 rad from the uniform one on all three domains, and we use 2.207 in the body. Every trained arm in Table 1 retains skill over it, including the weakest—the frozen no-augmentation arm on sunlamp, at 2.048 rad, keeps 0.15 rad. Translation is less comfortable: on sunlamp the constant scores 0.372, which two of the three augmented layer-4 seeds (0.405, 0.399) do not beat.

D Revision experiments

All runs in this appendix were added after review and pre-registered: predictions and decision rules were written down and committed before any result was opened. Selection uses synthetic validation only, as in the body.

Frozen-backbone block. Because the frozen backbone runs in inference mode without augmentation, its pooled features (and the ImageNet logits computed from them) can be cached once and heads retrained on them in minutes. Applying the published seed-42 head to the cache reproduces its published errors to within 0.0005. Table 6 gives three seeds (42, 43, 44) for each configuration, all with the published head, loss and selection rule. The 4,000-image cells use one fixed draw from the 28,000 (not the original 4,000-image set). The tuned head is the best of learning rate $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}\} \times \{20, 60\}$ epochs on seed 42, chosen on synthetic validation. The synthetic-BN head re-estimates every BatchNorm layer on the un-augmented synthetic training images (weights untouched) before caching.

Table 6: Frozen-backbone heads retrained on cached features: E_R (rad), mean $\pm$ s.d. over three seeds.

Configuration	Synthetic (val)	Lightbox	Sunlamp
4k images, 1000-d logits	1.651 ± 0.024	1.962 ± 0.012	2.040 ± 0.051
4k images, 2048-d pooled	1.395 ± 0.008	1.872 ± 0.016	2.062 ± 0.033
28k images, 1000-d logits	1.353 ± 0.026	1.878 ± 0.015	2.016 ± 0.027
28k images, 2048-d pooled (published config.)	1.030 ± 0.017	1.761 ± 0.026	1.986 ± 0.031
28k, pooled, tuned (lr 3×10^{-4}, 60 ep.)	0.929 ± 0.014	1.751 ± 0.035	2.001 ± 0.025
28k, pooled, synthetic-BN backbone	1.394 ± 0.002	2.064 ± 0.040	2.107 ± 0.010

Frozen backbone with augmentation. Augmented inputs cannot use the feature cache, so seeds 43 and 44 of the frozen augmented arm were trained end to end with the released code on an Apple M5 GPU; seed 42 is the published Colab run. Table 7 pairs each seed with the head-only retrain of the same seed in the published configuration (Table 6). Their conformal coverage is in App. B.

Table 7: Frozen backbone with augmentation, E_R (rad) per seed and paired difference from the head-only retrain without augmentation (same seed).

	E_R			ΔE_R vs. no augmentation		
Seed	Synth.	Lightbox	Sunlamp	Synth.	Lightbox	Sunlamp
42	1.180	1.718	1.864	$+0.169$	-0.029	-0.128
43	1.205	1.734	1.883	$+0.161$	-0.012	-0.069
44	1.169	1.673	1.881	$+0.134$	-0.117	-0.132

Augmentation ablation. Arm A is the full stack without blur and noise; arm B keeps only brightness/contrast jitter, gamma and random erasing (no glare blob, blur or noise). Both train layer 4 on seeds 42, 43 and 44 with the published configuration, and are read as per-seed differences. They were trained with the released code on an Apple M5 GPU, whereas the published arms were trained on Colab. A control rerun of the full stack (seed 42) on the M5 differs from its published counterpart by -0.053, +0.006 and -0.142 rad in E_R (synthetic, lightbox, sunlamp): within seed variation, but on sunlamp as large as the ablation effects themselves.

Test-time BatchNorm adaptation (AdaBN). For each layer-4 checkpoint we re-estimate only layer 4's BatchNorm statistics, the ones fitted on synthetic data, from the unlabelled images of one half of each real domain (a fixed random split by filename), leave every weight untouched, and evaluate on the other half (3,370 lightbox, 1,396 sunlamp images). Coverage uses each arm's published synthetic-calibrated $\hat{q}$. This is test-time adaptation: it uses target images, though not their labels. The frozen arms are excluded, as all their statistics are ImageNet's.

Table 8: Augmentation ablation, E_R (rad) per seed and paired difference from the published full stack (layer 4, same seed). Last column: sunlamp E_T.

		E_R			ΔE_R vs. full stack			Sunlamp
Arm	Seed	Synth.	Lightbox	Sunlamp	Synth.	Lightbox	Sunlamp	E_T
full stack	42	0.352	0.986	1.592	–	–	–	0.405
	43	0.302	0.997	1.354	–	–	–	0.399
	44	0.311	0.977	1.459	–	–	–	0.261
A: no blur/noise	42	0.320	1.070	1.408	-0.032	$+0.084$	-0.184	0.311
	43	0.328	1.001	1.360	$+0.026$	$+0.004$	$+0.006$	0.325
	44	0.320	1.081	1.342	$+0.009$	$+0.104$	-0.117	0.280
B: jitter, gamma, erasing	42	0.297	0.983	1.390	-0.055	-0.003	-0.202	0.286
	43	0.310	0.947	1.437	$+0.008$	-0.050	$+0.083$	0.309
	44	0.291	1.068	1.414	-0.020	$+0.091$	-0.045	0.330

Table 9: AdaBN on the held-out half of each real domain: unadapted $\rightarrow$ adapted. Coverage is of the 90% synthetic-calibrated rotation cone.

Aug.	Seed	Domain	E_R	E_T	Rotation coverage (%)
–	42	lightbox	1.171 $\rightarrow$ 1.074	0.202 $\rightarrow$ 0.149	21.1 $\rightarrow$ 23.9
		sunlamp	1.519 $\rightarrow$ 1.393	0.298 $\rightarrow$ 0.196	8.5 $\rightarrow$ 11.2
	43	lightbox	1.199 $\rightarrow$ 1.110	0.153 $\rightarrow$ 0.130	26.0 $\rightarrow$ 28.0
		sunlamp	1.644 $\rightarrow$ 1.422	0.351 $\rightarrow$ 0.199	8.8 $\rightarrow$ 14.5
	44	lightbox	1.160 $\rightarrow$ 1.057	0.156 $\rightarrow$ 0.132	25.6 $\rightarrow$ 29.3
		sunlamp	1.540 $\rightarrow$ 1.393	0.283 $\rightarrow$ 0.181	11.7 $\rightarrow$ 11.6
✓	42	lightbox	1.012 $\rightarrow$ 0.933	0.153 $\rightarrow$ 0.136	36.4 $\rightarrow$ 40.4
		sunlamp	1.594 $\rightarrow$ 1.400	0.406 $\rightarrow$ 0.210	12.3 $\rightarrow$ 18.6
	43	lightbox	1.013 $\rightarrow$ 0.908	0.165 $\rightarrow$ 0.170	32.2 $\rightarrow$ 35.5
		sunlamp	1.348 $\rightarrow$ 1.239	0.397 $\rightarrow$ 0.210	14.8 $\rightarrow$ 19.1
	44	lightbox	0.978 $\rightarrow$ 0.914	0.133 $\rightarrow$ 0.129	37.5 $\rightarrow$ 40.0
		sunlamp	1.483 $\rightarrow$ 1.342	0.262 $\rightarrow$ 0.192	14.6 $\rightarrow$ 19.6

Image statistics. Figure 1 compares pooled intensity densities, which cannot separate a soft bright blob from a hard edge. Table 10 adds spatial statistics, pooled over every image of each split after the same resize and grayscale conversion (augmentation seed 7, as in Figure 1): the Sobel gradient magnitude, scaled so a full black-to-white step reads 1; edge density, the share of pixels whose gradient exceeds 0.1; the clipped fraction, the share of pixels at or above $254/255$; and the total variation (TV) between pooled densities (48 intensity bins, 64 gradient bins). The intensity TV of the full stack reproduces Figure 1 (0.63, 0.80).

Table 10: Pooled image statistics and total-variation distance to each real domain.

	Edge	Clipped	TV intensity		TV gradient	
Pool	density (%)	(%)	lightbox	sunlamp	lightbox	sunlamp
lightbox	6.6	0.08	0.00	0.28	0.00	0.22
sunlamp	2.0	2.78	0.28	0.00	0.22	0.00
synthetic, no aug.	10.9	0.07	0.68	0.83	0.07	0.26
synthetic, full stack	10.7	1.96	0.63	0.80	0.29	0.51
synthetic, arm A	9.6	2.62	0.65	0.81	0.10	0.33
synthetic, arm B	9.7	1.67	0.66	0.82	0.09	0.31